

Microstrip Patch Antenna

Inset-Feed Optimization for High-Band Wireless Systems

PROJECT TEAM

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| MISSION OBJECTIVES



RESONANCE CONTROL

Achieving stable resonance at 5.0 GHz using parametric optimization of patch length (L).



IMPEDANCE MATCHING

Minimizing signal reflection through precise inset-feed depth (Fi) to hit the 50Ω locus.



BEAM OPTIMIZATION

Establishing a directive Broadside radiation pattern for point-to-point Wi-Fi 5 applications.

| ANTENNA STRUCTURAL ANATOMY

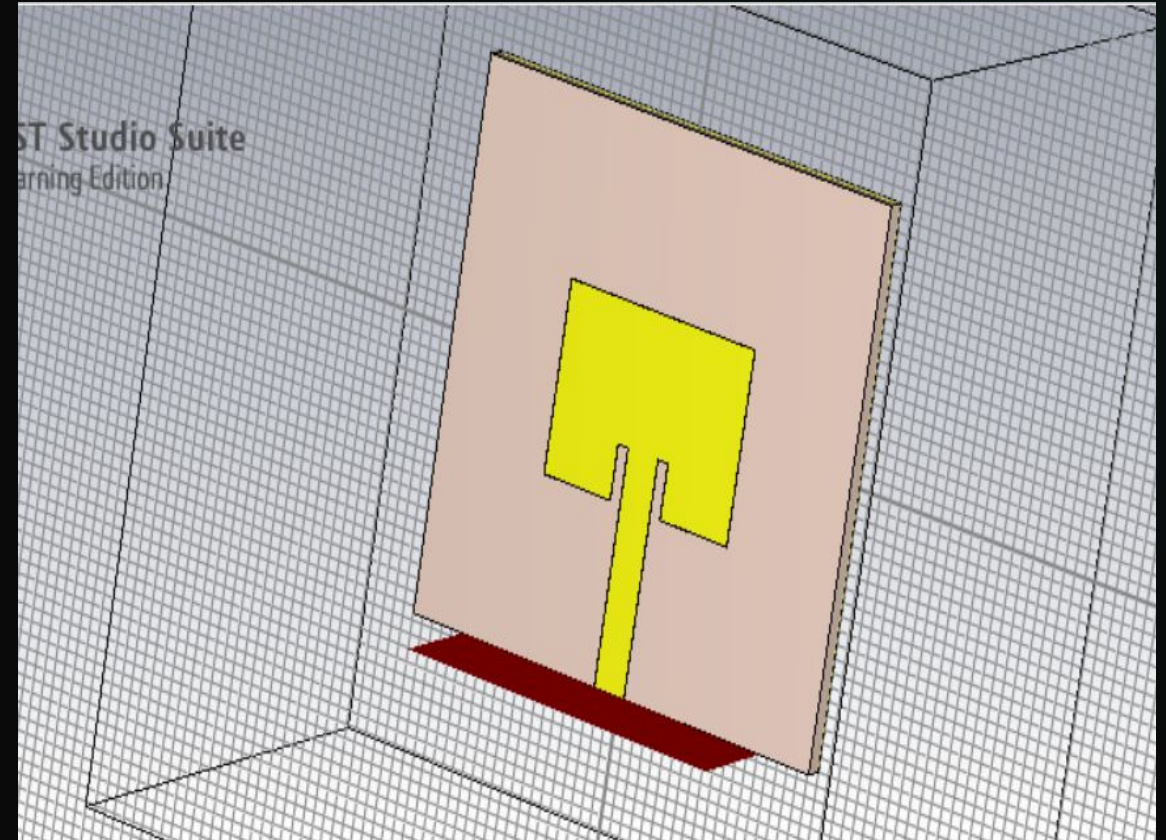
SUBSTRATE & GROUND PLANE

Layered sandwich architecture utilizing a high-conductivity radiator over a lossy dielectric.

■ **Patch:** Copper radiator etched for 5GHz.

■ **Substrate:** FR-4 ($h=1.6\text{mm}$, $\epsilon_r=4.3$).

— **Ground:** 40x40mm copper plane.



ENGINEERING PARAMETERS

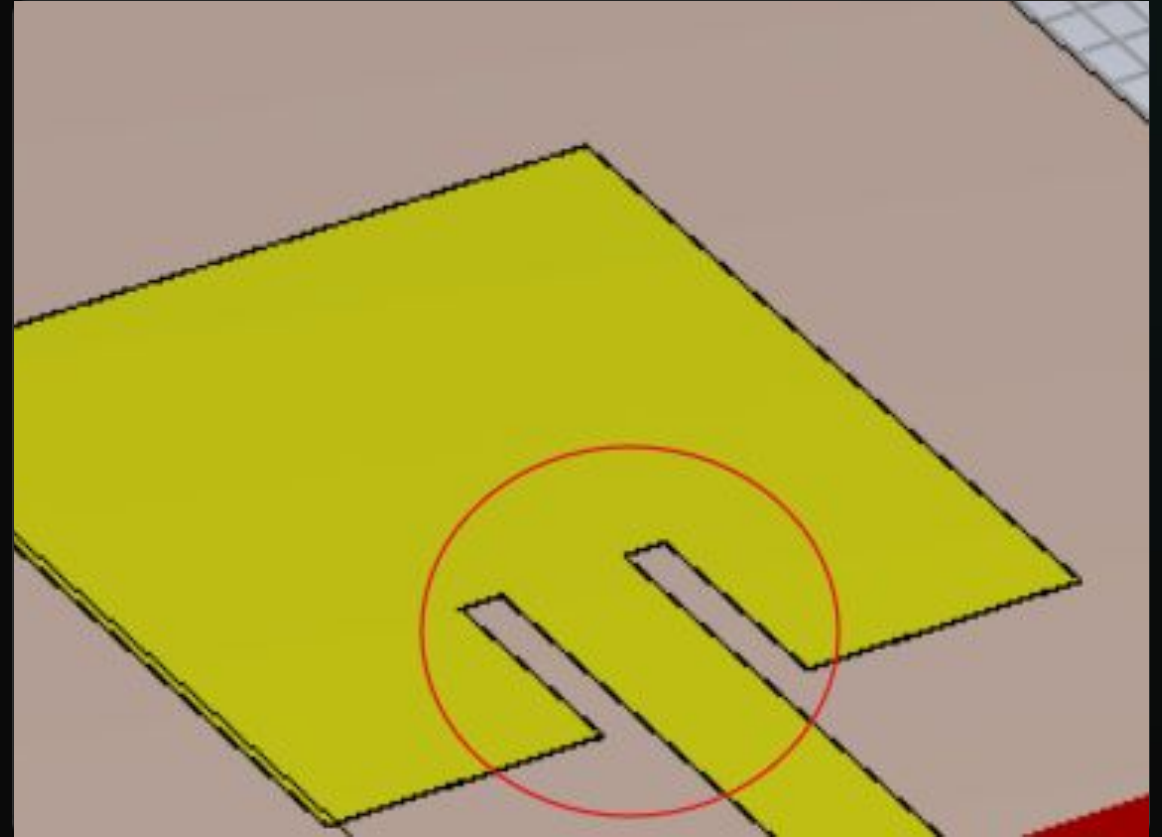
Parameter	Symbol	Value (mm)	Engineering Role
Patch Length	L	13.90	Frequency Resonator
Patch Width	W	18.40	Bandwidth & Impedance
Feed Width	Wf	3.00	50Ω Characteristic Line
Inset Depth	Fi	4.00	Locus Matching Depth
Feed Gap	G	1.00	Feeding Isolation Gap
Substrate Height	h	1.60	Dielectric Thickness
Copper Thickness	t	0.035	Conductive Layer Thickness
Ground L/W	Lsub / Wsub	40.00	Total Footprint Area

| THE INSET-FEED MATCHING

OVERCOMING EDGE IMPEDANCE

The patch edge exhibits high impedance ($\sim 250\Omega$). By cutting deep into the patch (F_i), we reach the 50Ω line impedance point.

$$Z_{in} = Z_{edge} \cos^2 \left(\frac{\pi}{L} F_i \right)$$

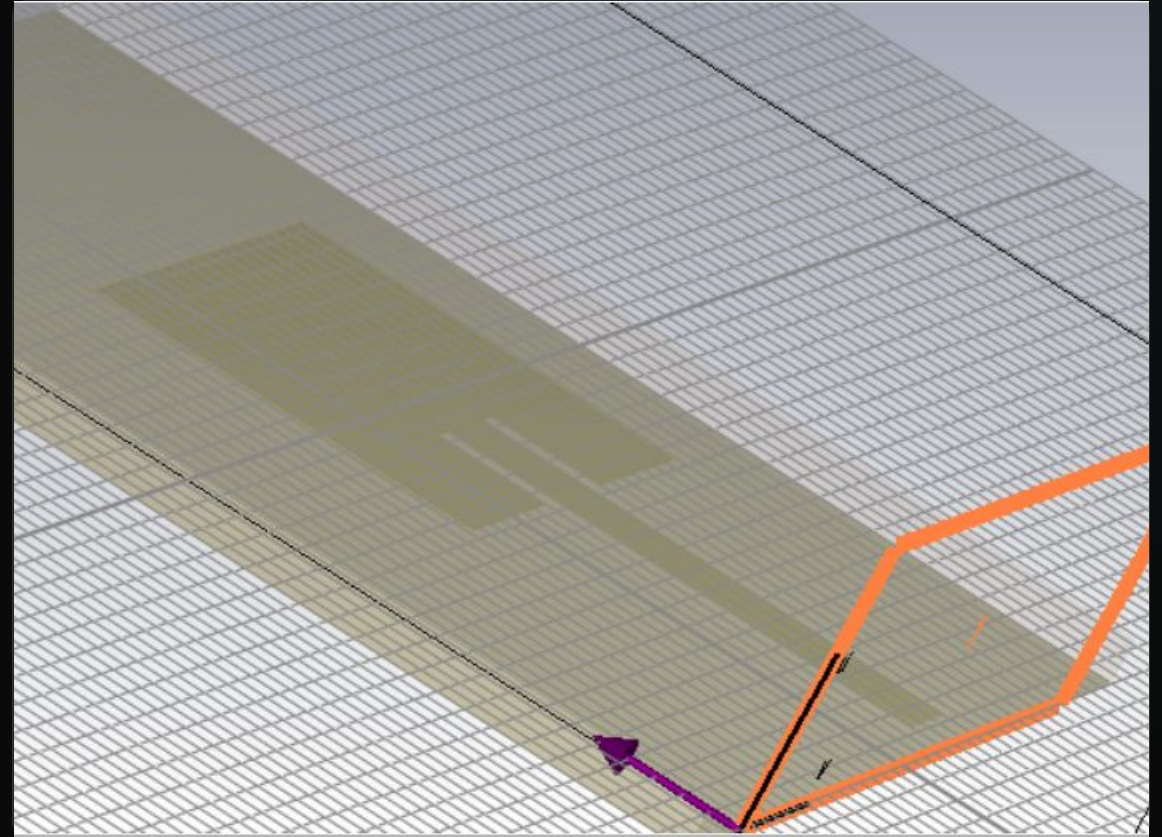


| PORT SETUP & EXCITATION

SIGNAL INJECTION

Port excitation was configured as a Waveguide Port to simulate the behavior of a 50 Ω SMA connector.

-  Port Multiplier: $(W+6h) \times (t+h)$.
-  Verified Line Impedance: 50.4 Ω .
-  Port centered on the microstrip feed.

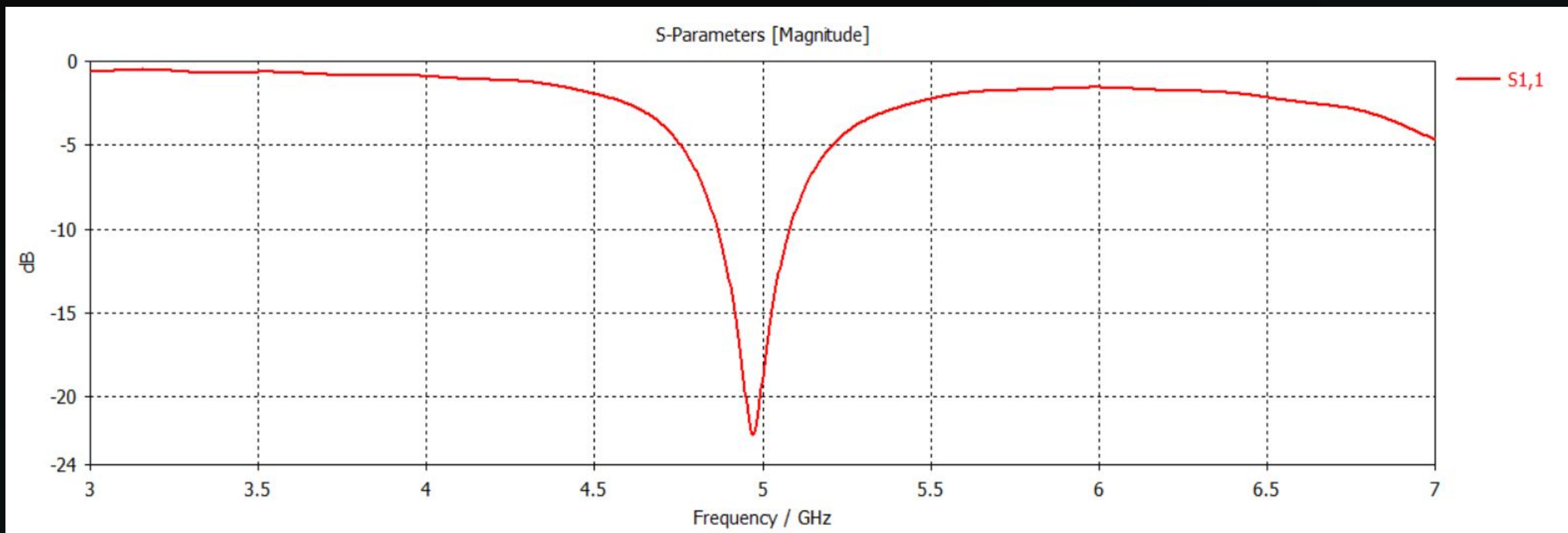


| S11 ANALYSIS

The reflection coefficient (S11) demonstrates the resonance quality.

Resonance: 4.972 GHz

Magnitude: -22.46 dB



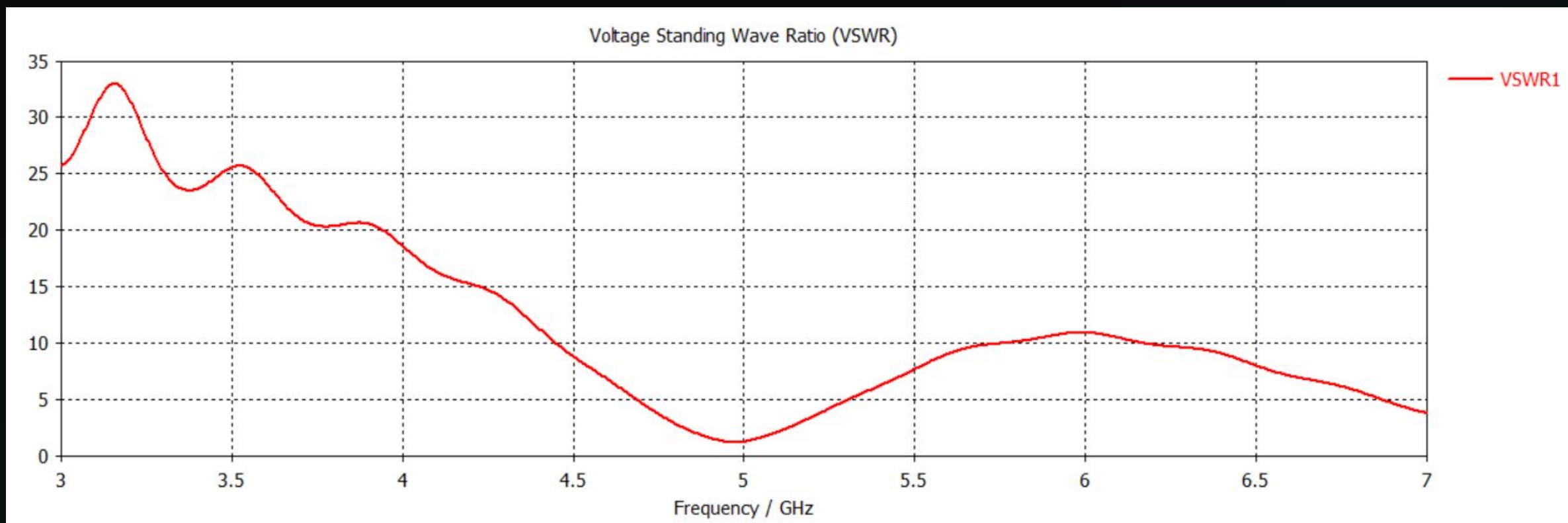
The sharp resonance indicates successful impedance matching.

VSWR ANALYSIS

Verification of standing wave ratio across the 3-7 GHz band.

Target: $VSWR < 2.0$

Result: 1.16 : 1



A VSWR of 1.16 indicates 99% power transmission efficiency.

| RADIATION PERFORMANCE

7.01 dBi

Peak Directivity

44%

Radiation Efficiency



Gain: Strong directional focus on the Broadside axis.



Efficiency: Affected by FR-4 loss tangent ($\tan\delta = 0.02$).

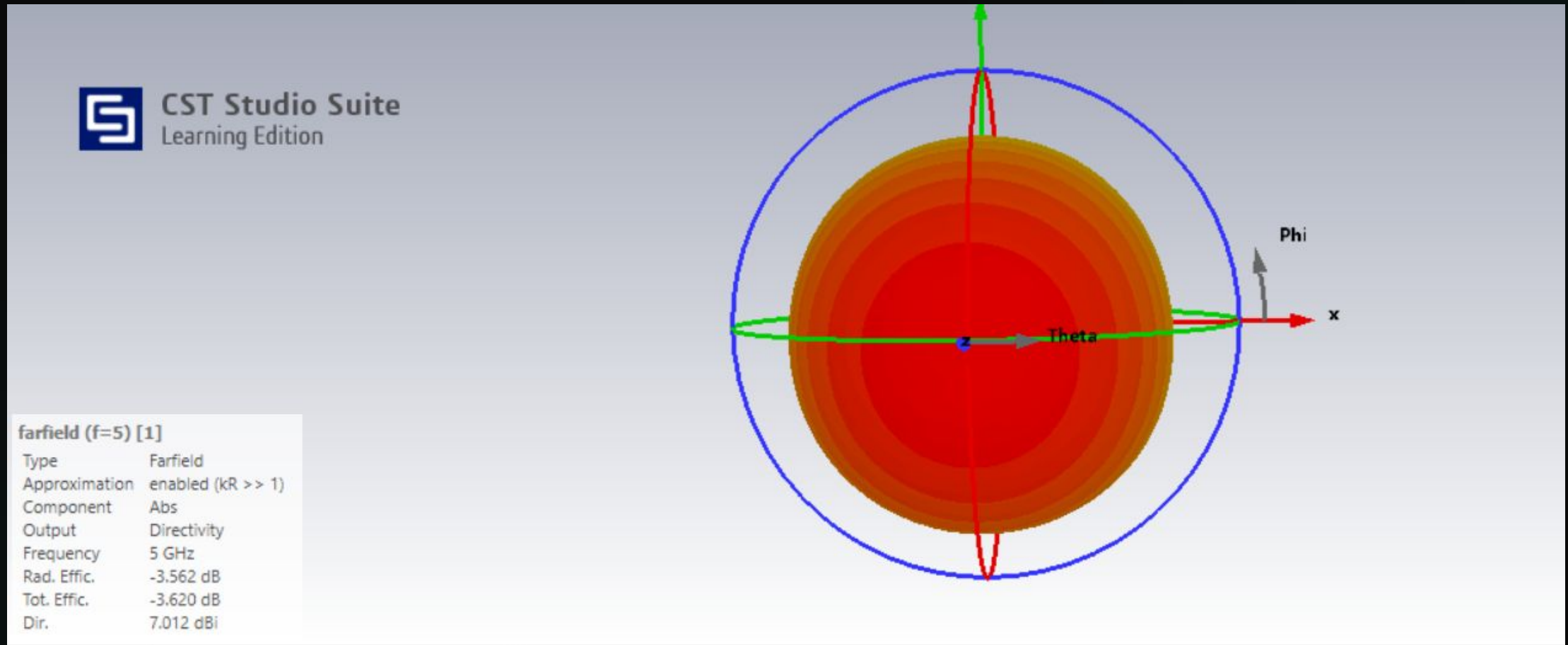


Total Gain: High gain relative to compact size.

FARFIELD PATTERN

Final far-field simulation shows energy concentration perpendicular to the patch plane.

Main lobe magnitude: 7.012 dBi at $\Phi=0$ deg.



3D FIELD VISUALIZATION

| PROJECT SUMMARY

- ✓ **Performance:** Successfully resonated at 4.97 GHz with -22 dB Return Loss.
- ✓ **VSWR:** Achieved an excellent ratio of 1.16, confirming matching success.
- ✓ **Engineering:** Inset-feed logic validated by high power efficiency.
- ✓ **Ready:** Parameterized design is stable and ready for fabrication.

Thank You